

Linear Algebra – MAT 2610

Section 2.3 (Characterizations of Invertible Matrices)

Dr. Jay Adamsson

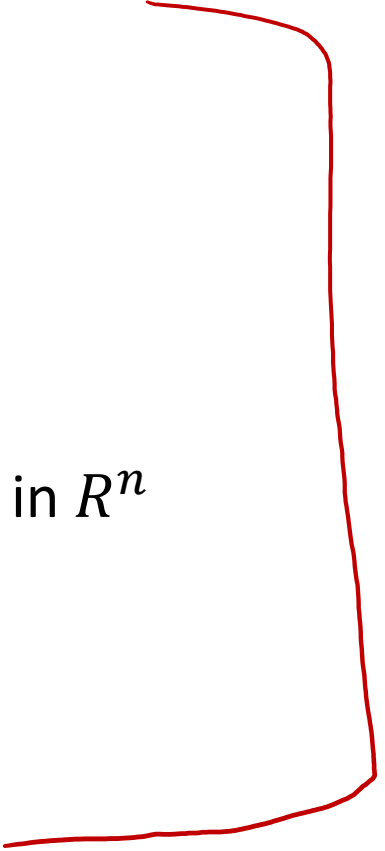
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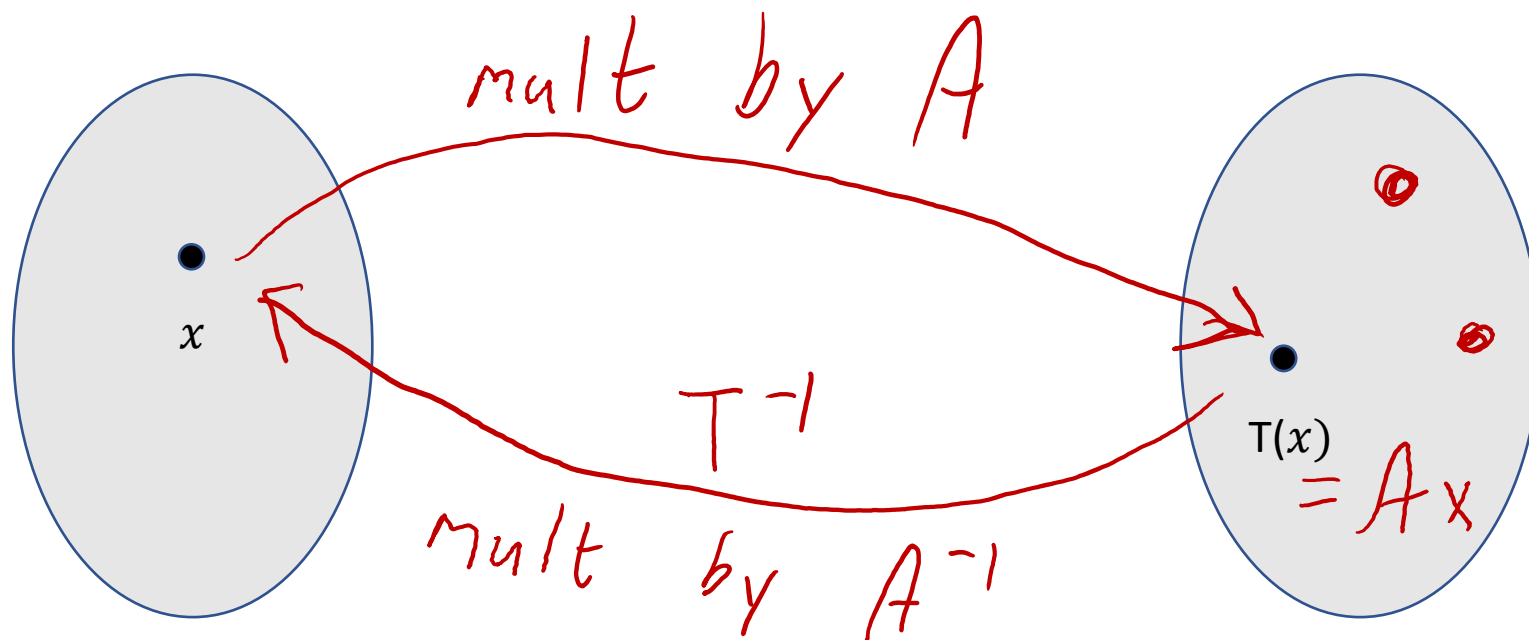
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Theorem 8 (Invertible Matrix Theorem)

If A is a square $n \times n$ matrix, then the following are equivalent:

- a. The matrix A is invertible
 - b. The matrix A is row equivalent to I_n
 - c. There are n pivot positions in A
 - d. The equation $Ax = 0$ has only the trivial solution
 - e. The columns of A are linearly independent
 - f. The linear transformation $x \mapsto Ax$ is one-to-one
 - g. The equation $Ax = b$ has at least one solution for every b in R^n
 - h. The columns of A span R^n
 - i. The linear transformation $x \mapsto Ax$ maps R^n onto R^n
 - j. There is an $n \times n$ matrix C such that $AC = CA = I_n$
 - k. The matrix A^T is invertible
- 



Theorem 9

Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a linear transformation and let A be the standard matrix for T . Then T is invertible if and only if A is an invertible matrix. In that case, the linear transformation S given by $S(x) = A^{-1}(x)$ is the unique function satisfying

- $S(T(X)) = x$ for all x in $\mathbb{R}^n$; and
- $T(S(X)) = x$ for all x in $\mathbb{R}^n$



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